

OPTIMAL POLICY FROM OPTIMAL VALUE FUNCTION

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Let us start with the definitions of Optimal Value Function and Optimal Policy (that we covered in the class on Markov Decision Processes).

Optimal State Value Function $V_*(s) = \max_{\pi} V_{\pi}(s)$ for all states $s \in \mathcal{S}$

Optimal Action-Value Function $Q_*(s, a) = \max_{\pi} Q_{\pi}(s, a)$ for all states $s \in \mathcal{S}$, for all actions $a \in \mathcal{A}$

π_* is an Optimal Policy if $V_{\pi_*}(s) \geq V_{\pi}(s)$ **for all policies** π and **for all states** $s \in \mathcal{S}$

Let us go beyond these formal definitions and develop an intuitive (and deeper) understanding of the above definitions. The definition of V_* says that for each state $s \in \mathcal{S}$, we go through all policies π and pick out the policy that maximizes $V_{\pi}(s)$. Because this maximization is done independently for each state $s \in \mathcal{S}$, presumably we could end up with different policies π that maximize $V_{\pi}(s)$ for different states. The definition of Optimal Policy π_* says that it is a policy that is “better than or equal to” (on the V_{π} metric) all other policies **for all** states (note that there could be multiple Optimal Policies). So the natural question to ask is whether there exists an Optimal Policy π_* that maximizes $V_{\pi}(s)$ **for all** states $s \in \mathcal{S}$, i.e., $V_*(s) = V_{\pi_*}(s)$ for all $s \in \mathcal{S}$. On the face of it, this seems like a strong statement. However, this answers in the affirmative. In fact,

Theorem 1. *For any Markov Decision Process*

- *There exists an Optimal Policy π_* , i.e., there exists a Policy π_* such that $V_{\pi_*}(s) \geq V_{\pi}(s)$ for all policies π and for all states $s \in \mathcal{S}$*
- *All Optimal Policies achieve the Optimal Value Function, i.e. $V_{\pi_*}(s) = V_*(s)$ for all $s \in \mathcal{S}$, for all Optimal Policies π_**
- *All Optimal Policies achieve the Optimal Action-Value Function, i.e. $Q_{\pi_*}(s, a) = Q_*(s, a)$ for all $s \in \mathcal{S}$, for all $a \in \mathcal{A}$, for all Optimal Policies π_**

Proof. First we establish a simple Lemma.

Lemma 1. *For any two Optimal Policies π_1 and π_2 , $V_{\pi_1}(s) = V_{\pi_2}(s)$ for all $s \in \mathcal{S}$*

Proof. Since π_1 is an Optimal Policy, from Optimal Policy definition, we have: $V_{\pi_1}(s) \geq V_{\pi_2}(s)$ for all $s \in \mathcal{S}$. Likewise, since π_2 is an Optimal Policy, from Optimal Policy definition, we have: $V_{\pi_2}(s) \geq V_{\pi_1}(s)$ for all $s \in \mathcal{S}$. This implies: $V_{\pi_1}(s) = V_{\pi_2}(s)$ for all $s \in \mathcal{S}$ $\square$

As a consequence of this Lemma, all we need to do to prove the theorem is to establish an Optimal Policy π_* that achieves the Optimal Value Function and the Optimal Action-Value Function. Consider the following Deterministic Policy (as a candidate Optimal Policy) $\pi_* : \mathcal{S} \rightarrow \mathcal{A}$:

$$\pi_*(s) = \arg \max_{a \in \mathcal{A}} Q_*(s, a) \text{ for all } s \in \mathcal{S}$$

First we show that π_* achieves the Optimal Value Function. Since $\pi_*(s) = \arg \max_{a \in \mathcal{A}} Q_*(s, a)$ and $V_*(s) = \max_{a \in \mathcal{A}} Q_*(s, a)$ for all $s \in \mathcal{S}$, π_* prescribes the optimal action for each state (that produces the Optimal Value Function V_*). Hence, following policy π_* in each state will generate the same Value Function as the Optimal Value Function. In other words, $V_{\pi_*}(s) = V_*(s)$ for all $s \in \mathcal{S}$. Likewise, we can argue that: $Q_{\pi_*}(s, a) = Q_*(s, a)$ for all $s \in \mathcal{S}$ and for all $a \in \mathcal{A}$.

Finally, we prove by contradiction that π_* is an Optimal Policy. So assume π_* is not an Optimal Policy. Then there exists a policy π and a state $s \in \mathcal{S}$ such that $V_\pi(s) > V_{\pi_*}(s)$. Since $V_{\pi_*}(s) = V_*(s)$, we have: $V_\pi(s) > V_*(s)$ which contradicts the definition of $V_*(s) = \max_\pi V_\pi(s)$

□